

DIVYANSH SINGH

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PROFESSIONAL SUMMARY

Aspiring AI/ML Engineer with hands-on experience in Machine Learning, Deep Learning, Explainable AI, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Python, Streamlit, and end-to-end AI application development. Built real-world projects including an Intelligent E-Commerce Customer Analytics Platform and a Dynamic AI Knowledge Assistant using FAISS, LangChain, Sentence Transformers, and Gemma LLM.

EDUCATION

Prof. Rajendra Singh (Rajju Bhaiya) University

Bachelor of Computer Applications (BCA)

Prayagraj

Aug 2024 – June 2027

HCL GUVI

Advanced AI & Machine Learning Program

PROJECTS

Intelligent E-Commerce Customer Analytics Platform

Python, Scikit-Learn, XGBoost, LightGBM, CatBoost, SHAP, Streamlit

- Integrated 9 relational e-commerce datasets into a unified analytics dataset.
- Performed data cleaning, preprocessing, feature engineering, and exploratory data analysis.
- Built customer segmentation using K-Means clustering.
- Developed purchase prediction models using Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost, LightGBM, and CatBoost.
- Implemented Customer Lifetime Value (CLV) prediction using regression models.
- Applied Explainable AI using SHAP Feature Importance, Summary Plot, Waterfall Plot, and Force Plot.
- Developed an interactive Streamlit dashboard for customer analytics and business insights.

Dynamic AI Knowledge Assistant using Automated Wikipedia Knowledge Base and Advanced RAG

Python, LangChain, FAISS, Sentence Transformers, Ollama, Gemma, Streamlit

- Developed an automated Wikipedia knowledge base generation pipeline using REST APIs.
- Performed metadata extraction, text cleaning, recursive, fixed, and semantic chunking.
- Generated embeddings using Sentence Transformers (all-MiniLM-L6-v2).
- Built FAISS vector databases for semantic retrieval.
- Implemented Retrieval-Augmented Generation (RAG) using LangChain.
- Integrated Gemma LLM through Ollama for context-aware question answering.
- Applied prompt engineering techniques to improve retrieval quality.
- Built a Streamlit application for AI-powered knowledge retrieval.

ADDITIONAL PROJECTS

- * **Content Monetization Modeler** – Python, Machine Learning, Regression
- * **Amazon Music Clustering** – Python, K-Means Clustering
- * **Comment Toxicity Detection** – Python, NLP, Machine Learning
- * **Ola Ride Insights** – Python, Data Analytics, Visualization
- * **Traffic Crash Analytics & Safety Intelligence Platform** – Python, Data Analysis, Streamlit

TECHNICAL SKILLS

Languages: Python, SQL

Machine Learning: Regression, Classification, Clustering, Feature Engineering, Model Evaluation, Ensemble Learning

Deep Learning: Neural Networks, TensorFlow, PyTorch

Generative AI: LLMs, RAG, LangChain, Prompt Engineering, Ollama, Gemma

NLP: Transformers, Sentence Transformers, Semantic Search, Embeddings

Explainable AI: SHAP, Feature Importance

Libraries: Pandas, NumPy, Scikit-Learn, XGBoost, LightGBM, CatBoost, Matplotlib, Plotly

Deployment & Tools: Streamlit, Git, GitHub, VS Code, Jupyter Notebook, FAISS

CERTIFICATIONS

Advanced AI & Machine Learning Program

HCL GUVI

ACHIEVEMENTS

- Built two end-to-end AI projects in Machine Learning and Generative AI.
- Implemented Explainable AI using SHAP.
- Developed Retrieval-Augmented Generation pipelines using LangChain and FAISS.
- Built interactive Streamlit dashboards for AI applications.